COGNIFYZ IT SOLUTIONS

Machine Learning Internship

TASK 2

Restaurant Recommendation System

Content-Based Filtering Using User Preferences

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Internship Domain: Machine Learning

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Based on the completed Task 2 notebook and Cognifyz Machine Learning task requirements.

1. Objective

The objective of this task is to create a restaurant recommendation system based on user preferences. The implemented system uses a content-based approach: relevant restaurant attributes are preprocessed and represented numerically, user preferences are represented in the same feature space, and restaurants are identified according to the selected cuisine, price range, and city. Matching restaurants are then ranked using Aggregate Rating and Votes.

2. Task Requirements and Scope

The Cognifyz Task 2 specification requires the following core activities:

- Preprocess the dataset by handling missing values and encoding categorical variables.
- Determine suitable criteria for restaurant recommendations, such as cuisine preference and price range.
- Implement a content-based filtering approach using the preferred criteria.
- Test the recommendation system with sample user preferences and evaluate recommendation quality.

The implementation in this report follows those requirements. The core matching criteria selected in the project are Cuisine, Price Range, and City. Online Delivery and Table Booking are included as user-input fields and displayed in the results, but the current recommendation filter does not use them as active matching conditions.

3. Dataset Description

The project uses the restaurant dataset stored as Dataset.csv. The completed notebook reports 9,551 restaurant records and 21 original columns.

Dataset Item	Observed Value
Rows	9,551
Original columns	21
Unique cities	141
Price ranges	1, 2, 3, 4
Missing Cuisines before handling	9

Selected recommendation-related fields include Restaurant Name, Cuisines, Price range, City, Has Online delivery, and Has Table booking. Aggregate rating and Votes are later added as ranking-support fields.

4. Data Preprocessing

4.1 Missing-Value Handling

A missing-value check was performed on the recommendation features. The Cuisines column contained 9 missing values; these were replaced with the label “Unknown”. After replacement, no missing values remained in Cuisines.

```
df["Cuisines"] = df["Cuisines"].fillna("Unknown")
```

4.2 Feature Selection

Only the fields relevant to recommendation matching were retained. Restaurant Name was preserved for displaying recommendations, while the matching features were Cuisines, Price range, City, Has Online delivery, and Has Table booking.

4.3 Cuisine Preparation and Encoding

Because a restaurant may contain multiple cuisines in the same record, cuisine strings were split on commas and whitespace was removed. MultiLabelBinarizer was then used to create 146 binary cuisine features across the 9,551 restaurants.

```
from sklearn.preprocessing import MultiLabelBinarizer
mlb = MultiLabelBinarizer()
cuisine_encoded = mlb.fit_transform(X_recommendation["Cuisines"])
```

4.4 City Encoding

City is a single categorical feature, so OneHotEncoder was used. The dataset produced 141 city features.

4.5 Binary Feature Encoding

Has Online delivery and Has Table booking contain only Yes/No values. They were mapped to 1/0 so they could be included in the numerical feature representation.

5. Restaurant and User Feature Representation

The encoded feature blocks were combined into a single restaurant feature matrix. The final restaurant representation contains 290 features per restaurant: 146 cuisine features + 141 city features + 1 price feature + 1 delivery feature + 1 booking feature.

A sample user preference was also converted into the same 290-feature representation. For the main demonstration, the user selected Japanese cuisine, Price Range 3, Makati City, Online Delivery = No, and Table Booking = Yes.

6. Content-Based Filtering and Recommendation Logic

Cosine similarity was first calculated between the user feature vector and all 9,551 restaurant feature vectors. This produced one similarity score for each restaurant. The initial similarity

ranking was useful for exploration, but the final recommendation function applies the core user constraints directly.

The final recommendation process is:

- Check whether the requested cuisine is present in the restaurant cuisine list.
- Require the restaurant price range to equal the user-selected price range.
- Require the restaurant city to equal the user-selected city.
- Sort the matching restaurants by Aggregate Rating in descending order, using Votes as the secondary tie-breaker.

7. Recommendation Function

The workflow was wrapped into a reusable function named recommend_restaurants(), allowing different user preferences to be tested without rewriting the recommendation logic.

```
def recommend_restaurants(cuisine, price_range, city, online_delivery, table_booking):
    # filter by cuisine, price range, and city
    # rank by Aggregate rating and Votes
    # return the recommended restaurant fields
```

The function currently accepts Online Delivery and Table Booking as inputs and displays them, but these two fields are not used as active filters in the current implementation. This is stated explicitly so that the report matches the notebook behavior.

8. Testing and Results

Three sample preference tests were performed using values observed in the dataset.

Test	Cuisine	Price	City	Matching restaurants	Validation
Test 1	Japanese	3	Makati City	2	Cuisine: True; Price: True; City: True
Test 2	Chinese	1	Albany	2	Cuisine: True; Price: True; City: True
Test 3	Indian	2	New Delhi	948	Cuisine: True; Price: True; City: True

8.1 Test Case 1 - Japanese

User preference: Japanese cuisine, Price Range 3, Makati City. The system identified 2 core matches. The final ranking placed Le Petit Souffle first with an Aggregate Rating of 4.8 and Izakaya Kikufuji second with a rating of 4.5.

Restaurant	Cuisine	Rating	Votes
Le Petit Souffle	French, Japanese, Desserts	4.8	314

Izakaya Kikufuji	Japanese	4.5	591

8.2 Test Case 2 - Chinese

An initial Chinese + Price 1 + Makati City test returned 0 matches. This demonstrated that an exact combination may not exist in the dataset. A valid Chinese + Price 1 + Albany test was then selected from the actual dataset and returned 2 matching restaurants.

8.3 Test Case 3 - Indian

User preference: Indian cuisine, Price Range 2, New Delhi. The system returned 948 restaurants matching the three core criteria. Validation confirmed that the returned results satisfied the requested cuisine, price range, and city conditions.

9. Recommendation Validation

Programmatic validation was applied to all three test cases. For each test, the returned recommendations were checked for cuisine, price-range, and city consistency. All three validation checks returned True for each tested criterion.

Test case	Cuisine match	Price match	City match
Test 1	True	True	True
Test 2	True	True	True
Test 3	True	True	True

10. Results and Discussion

- The system successfully preprocesses the selected categorical and binary features and produces a consistent 290-dimensional restaurant representation.
- The same feature space is used to represent user preferences, enabling content-based similarity calculation.
- The final recommendation function respects the selected cuisine, price range, and city conditions.
- Ranking by Aggregate Rating, with Votes as a secondary criterion, provides a simple and explainable ordering among matching restaurants.
- The number of recommendations varies substantially by preference combination; an exact combination can yield no matches, while broader combinations such as Indian + Price 2 + New Delhi yield many matches.

11. Limitations and Future Improvements

The current implementation is a focused internship-level content-based recommender and has several clear limitations:

- Online Delivery and Table Booking are not yet incorporated into the final filtering logic even though they are accepted as function inputs.
- The current function returns all restaurants satisfying the core criteria rather than imposing a final Top-N limit in every case.
- The current similarity calculation is demonstrated as part of the content-based workflow, while the final ranking relies on direct preference matching followed by rating and vote ranking.
- Future versions could use weighted feature similarity, service-preference filtering, Top-N controls, normalized rating/vote signals, or a user-facing application interface.

12. Conclusion

Task 2 was completed by developing a content-based restaurant recommendation workflow using user-selected cuisine, price range, and city preferences. The dataset was cleaned, categorical attributes were encoded, user and restaurant feature representations were aligned, and recommendation quality was checked across multiple sample cases. The validation results confirm that the final recommendation function returns restaurants consistent with the selected core criteria.

13. Submission Readiness Checklist

Item	Status / Note
Task 2 notebook	Completed and tested from top to bottom
Source code	Included in the notebook
Recommendation outputs	Present in notebook test cells
Testing and validation	3 sample tests with validation
Documentation/report	This PDF report
Project demonstration video	Optional according to the internship email
ZIP submission	Combine notebook, dataset if permitted/required, report, and other required outputs into one ZIP for the official submission form

Reference basis: Cognifyz Technologies Machine Learning Internship Task 2 specification and the completed Task_2_Restaurant_Recommendation notebook. The report intentionally describes only behavior and results supported by those materials.